Clothing Size Recommendation Using Body Measurement Estimation

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Abstract. Incorrect apparel sizing in online shopping leads to high return rates, customer dissatisfaction, and environmental waste. Traditional methods, such as manual measurements, are impractical for remote shoppers. This project presents the clothing size recommendation system, which requires users to input their height and capture two images (front and side) with guided on-screen overlays to ensure proper alignment. The system employs YOLOv11 model to extract key body points from both images, which, when combined with the user's height through scaling techniques, yield accurate estimations of 3D body dimensions such as shoulder width and belly circumference. These measurements are then fed into multiple machine learning models for size classification. After comprehensive evaluation of four different algorithms—Logistic Regression, Random Forest, Support Vector Machine, and XGBoost—we identified Random Forest as the optimal model, achieving 97.50% accuracy in clothing size prediction.

Keywords: Clothing size recommendation · Body measurement estimation · Computer vision · Pose estimation · YOLOv11 · Random Forest

1 Introduction

Clothing size is a big problem for many people, especially when online shopping or those who do not know about their body size, which leads to incorrect clothes size choices, and increases the return rate of many online shops. Some traditional sizing calculation methods, such as manual measurements, often fail to capture the unique variations in individual body shapes, and it is hard to do it by themselves if they are online shopping [1] [2]. In order to struggle with these challenges, this paper introduces a clothing size recommendation system that uses a multi-image method to estimate body measurements, aiming to deliver more precise and personalized size recommendations.

The proposed system allows users to input their height and capture two images—one from the front and one from the side—using guided on-screen overlays to ensure proper alignment and consistency. This dual-view capture addresses the limitations of single-image approaches by providing complementary perspectives that are essential accurately estimating key body dimensions such as shoulder width, belly circumference, and other critical measurements [3]. By integrating advanced computer vision techniques, specifically through the use of the YOLOv11 library for pose estimation, the system extracts 2D landmarks from both views. After that, the body measurements are input into multiple machine learning models to determine the most suitable classification approach for size recommendation.

The enhanced measurement accuracy achieved through the multi-image approach is pivotal in bridging the gap between the simplicity of automated digital methods and the precision of traditional manual measurements. Once the body dimensions are estimated, they are processed through a comprehensive model comparison study involving four different machine learning algorithms: Logistic Regression, Random Forest, Support Vector Machine (SVM), and XGBoost. This comparative analysis ensures the selection of the most effective model for mapping body measurements to appropriate clothing sizes.

In this study, we describe the methods used for image capture and processing, show the whole design and implementation of the clothing size suggestion system, present a thorough comparison of multiple machine learning models, and assess the system's effectiveness in real world problems.

This project's main contributions are:

- The creation of an intuitive user interface that makes it easier to take precise front and side view photos.
- The use of pose estimation techniques to extract important body landmarks.
- The reconstruction of 3D body measurements using height-based scaling.
- A comprehensive comparison of four machine learning models for clothing size classification.
- The identification of Random Forest as the optimal model for size prediction with 97.50% accuracy.
- The integration of these measurements with an extensive database of clothing sizes to provide customized size recommendations.

The rest of this paper has the following structure: Section 2 explains the project architecture and frameworks that we use, including image processing, human pose detection, and calculating body size method. The following section represents our results, evaluating the measurement accuracy and limitations. Section 4 discusses some improvements that we planned and outlines some future work to increase the accuracy of both body size measurement process and size recommendation process. The final section gives the conclusion of our paper by summarizing all the contents and our effort in this project, also discussing the impact of this for the online fashion retail industry.

2 Related Work

In this section, we review recent research in three major areas that are relevant to our study: YOLO-based human detection and pose estimation, logistic regression and classical statistical modeling, and clothing size estimation for fashion applications. Each of these research streams provides insights that contribute to the development of automated sizing systems, while also presenting limitations that motivate our proposed approach.

2.1 YOLO-Based Human Detection and Pose Estimation

The YOLO family of models has become a cornerstone for real-time object detection due to its balance between speed and accuracy. In recent years, multiple improvements have been proposed to enhance YOLO for challenging environments. For example, CST-YOLO introduced convolutional and transformer-based modules to boost detection performance for small-scale objects [4], while lightweight versions such as YOLOv7-tiny [5] demonstrated the feasibility of deploying detection models on resource-constrained devices without sacrificing significant accuracy. Other works, such as the improved YOLOv7 for obscured pedestrian detection [6] and YOLO-RSFM [7], extended the model with attention mechanisms and transformer decoder heads to improve robustness against occlusion and scale variations.

These studies highlight YOLO's flexibility and its potential beyond conventional object detection. However, most research has focused on general-purpose detection tasks rather than directly estimating anthropometric measurements. In our work, we leverage YOLOv11 not only for object detection but specifically for pose landmark extraction, enabling the computation of precise body measurements from multi-view images.

2.2 Logistic Regression and Statistical Modeling

Parallel to advancements in deep learning, logistic regression continues to serve as a reliable statistical method for classification tasks. Its interpretability and efficiency make it an attractive choice, especially when categorical outputs are required. Recent works have investigated ways to enrich logistic regression models with additional data and features, thereby improving their robustness and predictive capacity [8]. Other studies have emphasized model selection strategies [9], comparing criteria such as AIC and BIC, and exploring the trade-off between model complexity and generalization performance. Furthermore, in the fashion domain, regression-based approaches have been employed to predict missing body measurements from partial inputs, demonstrating the continued relevance of statistical models [10].

While these works highlight the versatility of logistic regression, most applications remain either theoretical or focused on general data problems. Few have considered its integration with computer vision-based feature extraction pipelines. In contrast, our approach employs logistic regression as the final mapping stage, translating continuous anthropometric features extracted from images into discrete clothing size categories.

2.3 Clothing Size Estimation and Fashion Applications

The fashion industry has also seen growing interest in body size prediction and clothing recommendation systems. One line of research focuses on 2D image-based estimation. For instance, CNN-based T-shirt size prediction systems [11] have explored body contour detection combined with keypoint estimation, achieving moderate accuracy but struggling with variations in pose, camera distance, and clothing texture. Another direction relies on advanced 3D body scanning technologies [12], which allow highly accurate measurements and clustering of body shapes into size groups. Although precise, these methods require specialized equipment, limiting their scalability in consumer applications.

Recent works have attempted to bridge this gap by exploring hybrid approaches, such as combining pose estimation with limited anthropometric data, yet accuracy often falls short of the standards needed for reliable e-commerce deployment. The tension between practicality (2D smartphone images) and accuracy (3D scanning systems) remains an open challenge in this domain.

2.4 Summary

Overall, prior research demonstrates the individual strengths of YOLO-based detection models, logistic regression, and specialized size estimation systems. However, these approaches often operate in isolation: YOLO is rarely applied to measurement estimation, logistic regression is usually studied in abstract settings, and size estimation methods either rely on costly hardware or suffer from limited accuracy in 2D environments. Our work contributes by integrating these streams into a unified pipeline. Specifically, we adopt YOLOv11 for robust multi-view pose landmark extraction and apply logistic regression to map estimated body dimensions to standardized clothing sizes. This design achieves a practical balance between accuracy and accessibility, offering a scalable solution for real-world online shopping platforms.

3 Background

The rapid growth of online clothing shopping has led to a need for automated systems that measure the human body size of customers to recommend the most suitable clothing size. We propose an approach based on knowledge in the fields of computer vision, human pose estimation, and statistical modeling, and combine image processing libraries such as YOLOv11 and OpenCV to simplify the measurement process.

3.1 Core Definitions and Concepts

Human Pose Detection This is a part of computer vision that involves recognizing and roughly locating the major joints of the body (e.g., shoulders, elbows, waist, etc.) from images or video streams.

Pose generation techniques, such as those implemented in YOLOv11, use convolutional neural networks (CNNs) to predict a set of heatmaps or keypoint maps, each corresponding to one anatomical landmark. By training on large datasets of annotated human poses, the network learns to extract spatial features—like limb orientation, joint appearance, and body silhouette—and to output precise coordinate estimates for each joint in real time.

These techniques are useful because:

- **Real-time feedback:** The high processing speed of single-stage CNNs (one forward pass per frame) enables applications such as live fitness coaching, gesture-based interfaces, and interactive gaming without perceptible lag.
- **Robustness to variation:** Deep convolutional filters automatically learn to handle different viewpoints, clothing, lighting, and partial occlusions, making pose detection reliable across diverse environments.
- **Downstream analytics:** Exact joint coordinates allow computation of angles, distances, and movement trajectories, which power higher-level tasks like activity recognition, ergonomic assessment, rehabilitation monitoring, and sports performance analysis.
- **Personalization:** In our clothing-size recommendation system, precise measurements (e.g., shoulder width, arm length) extracted from live video ensure that suggested sizes match each user's unique body proportions, reducing returns and improving customer satisfaction.

YOLOv11 YOLO (short for "You Only Look Once") is a real-time object detection system that uses convolutional neural networks to detect and localize objects in images and video streams. It was originally introduced by Joseph Redmon and has since been further developed by the computer vision community. YOLO is well known for its speed and efficiency, making it suitable for real-time applications on both high-end devices and mobile platforms [13]. Although YOLO was primarily designed for object detection, its architecture can be adapted for other tasks such as human pose estimation. In our application, a variant of YOLO is employed to automatically extract precise body measurements from a live camera stream, which then serve as the basis for generating personalized clothing size recommendations.

Logistic Regression Logistic regression is a core statistical classification method that models the probability of a categorical outcome given one or more predictor variables. In our system, we use it to map continuous body measurements (e.g., shoulder width, waist circumference) into discrete clothing size categories (Small, Medium, Large, etc.).

Binary Logistic Function. For the simplest case of two classes (e.g. "Medium" vs. "Not Medium"), the model assumes

$$P(Y = 1 \mid \mathbf{x}) = \sigma(z) = \frac{1}{1 + e^{-z}}, \tag{1}$$

where

$$z = \beta_0 + \beta_1 x_1 + \cdots + \beta_n x_n, \tag{2}$$

where

- $\mathbf{x} = (x_1, \ldots, x_n)$ is the feature vector of measured dimensions.
- β_0 is the intercept (bias) term, shifting the decision boundary.
- β_i is the weight corresponding to feature x_i, indicating how strongly x_i influences the log-odds.
- $\sigma(\cdot)$ is the logistic (sigmoid) function, which "squashes" real valued z into the interval $(0, 1)$.

We interpret $P(Y = 1 \mid \mathbf{x}) > 0.5$ (or another chosen threshold) as predicting class "1."

Multinomial Extension. To handle $K > 2$ size categories simultaneously, we generalize to

$$P(Y = k \mid \mathbf{x}) = \frac{\exp\left(\beta_{k0} + \boldsymbol{\beta}_k^T \mathbf{x}\right)}{\sum_{j=1}^{K} \exp\left(\beta_{j0} + \boldsymbol{\beta}_j^T \mathbf{x}\right)}, \quad k = 1, \ldots, K, \tag{3}$$

where each class k has its own parameter vector $(\beta_{k0}, \boldsymbol{\beta}_k)$. The predicted size is the k with the highest posterior probability.

Parameter Estimation. All parameters $\{\beta_{k0}, \boldsymbol{\beta}_k\}$ are learned by maximizing the regularized log-likelihood on a labeled training set:

$$\mathcal{L}(\boldsymbol{\beta}) = \sum_{i=1}^{N} \sum_{k=1}^{K} \mathbb{I}(y^{(i)} = k) \ln P(Y = k \mid \mathbf{x}^{(i)}) - \frac{\lambda}{2} \sum_{k=1}^{K} \|\boldsymbol{\beta}_k\|_2^2, \tag{4}$$

where λ controls the strength of L_2 regularization to prevent overfitting.

Interpretation of Parameters.

- A positive coefficient β_{ki} means that increasing feature x_i raises the log odds of class k relative to the baseline.
- The magnitude $|\beta_{ki}|$ reflects how strongly x_i influences the probability for size k.
- The intercept β_{k0} shifts the model's bias toward or away from class k when all $x_i = 0$ (after normalization).

In our application, the input $\mathbf{x}$ is the scaled set of body measurements produced by YOLOv11, and the trained logistic model outputs—for each size category—the probability that the measurements belong to that category. We then recommend the size with maximum probability, optionally enforcing a minimum confidence threshold (e.g. 0.7) to prompt the user for retakes if the classification is uncertain.

Random Forest Random Forest [14] is an ensemble learning algorithm that builds a collection of decision trees and aggregates their predictions to improve generalization. Each tree is trained on a bootstrap sample of the training data, and at each split only a random subset of features is considered. This randomness reduces correlation between trees, leading to lower variance compared to a single decision tree.

For classification, the final prediction is obtained by majority voting across all trees:

$$\hat{y} = \text{mode}\{h_1(\mathbf{x}), h_2(\mathbf{x}), \ldots, h_T(\mathbf{x})\}, \tag{5}$$

where $h_t(\mathbf{x})$ is the class prediction of tree t, and T is the total number of trees.

Random Forests are robust to noise, can capture nonlinear feature interactions, and naturally handle multi-class classification. In our context, they provide a strong nonparametric baseline for mapping anthropometric features into clothing size categories.

Support Vector Machine (SVM) Support Vector Machine [15] is a margin-based classifier that seeks a hyperplane separating classes with maximum margin. For linearly separable data, the decision function is

$$f(\mathbf{x}) = \text{sign}\big(\mathbf{w}^T\mathbf{x} + b\big), \tag{6}$$

where $\mathbf{w}$ is the weight vector orthogonal to the hyperplane and b is the bias. The optimization maximizes the margin $\frac{2}{\|\mathbf{w}\|}$ subject to classification constraints.

To handle nonlinearity, kernel functions $\kappa(\mathbf{x}_i, \mathbf{x}_j)$ map inputs into a higher-dimensional space, allowing separation by a linear hyperplane there. Common kernels include polynomial and radial basis function (RBF). For multi-class size classification, strategies such as one-vs-one (OvO) or one-vs-rest (OvR) are applied.

SVMs are effective with high-dimensional data and small training sets, making them suitable for exploring the discriminative power of body measurements beyond logistic regression.

Extreme Gradient Boosting (XGBoost) XGBoost [16] is a scalable, high-performance implementation of gradient boosted decision trees. It builds an additive model of M trees, where each new tree f_m is trained to correct the residual errors of the ensemble so far:

$$\hat{y}^{(i)} = \sum_{m=1}^{M} f_m(\mathbf{x}^{(i)}), \quad f_m \in \mathcal{F}, \tag{7}$$

with $\mathcal{F}$ the space of regression trees. The objective combines a differentiable loss function ℓ and a regularization term Ω:

$$\mathcal{L} = \sum_{i=1}^{N} \ell\big(y^{(i)}, \hat{y}^{(i)}\big) + \sum_{m=1}^{M} \Omega(f_m). \tag{8}$$

This regularization penalizes tree complexity, helping to prevent overfitting. XGBoost uses second-order gradient information for more accurate optimization and incorporates techniques like shrinkage, subsampling, and column sampling for better generalization.

For our clothing size prediction task, XGBoost offers state-of-the-art accuracy in tabular classification, effectively capturing complex interactions among anthropometric features.

Metrics for evaluate Model Metrics refer to a set of numbers that provide information about a particular process or activity. They are essential for measuring performance and success in various fields, including business and data analysis. Metrics help organizations track progress, make informed decisions, and improve strategies for growth.

– **Precision:** the proportion of correctly predicted positive samples out of all predicted positive samples. A high precision means fewer false positives, meaning the model shows a confidence in predictions.

$$Precision = \frac{TP}{TP + FP} \tag{9}$$

– **Recall:** the proportion of correctly predicted positive samples out of all actual positive samples. A high recall means fewer false negatives, indicating the model detects most positive cases but may include some incorrect ones.

$$Recall = \frac{TP}{TP + FN} \tag{10}$$

– **F1-score:** the harmonic mean of Precision and Recall. A high F1-score requires both Precision and Recall to be high.

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{11}$$

– **Weighted-Averaged F1-score:** a variant of the F1-score that takes class imbalance into account by weighting each class's F1-score according to its number of true instances in the dataset.

$$F1_{\text{weighted}} = \sum_{i=1}^{3} w_i F1_i \tag{12}$$

– **Accuracy:** The proportion of correctly predicted samples out of all samples. While useful, accuracy alone may be misleading for imbalanced datasets.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{13}$$

**Note: TP - True positive, FP - False Positive, FN - False Negative*

3.2 Existing Theories

Statistical Modeling with Logistic regression Logistic regression provides a simple yet powerful tool for predicting unknown measurements based on known variables. In our system, it is used to classify body measurements into discrete size categories by leveraging the relationship between variables such as height and weight and categorical measures like chest or waist circumference. This approach enables us to deliver fast and reliable size recommendations.

4 Proposed Approach

4.1 User's size estimation

Human key points detection To detect human key points, we use the YOLOv11 pretrained model, which is specifically designed for human pose estimation. This model is modified to identifying and localizing key body such as the head, shoulders, elbows, wrists, hips, knees, and ankles from 2D images or video frames. In our approach, the YOLOv11 model processes input images and returns the coordinates of key points after capturing 2 images of user including front view and side view. These coordinates are essential for determining body posture, extracting body proportions, and further calculating measurements for our clothing size recommendation system.

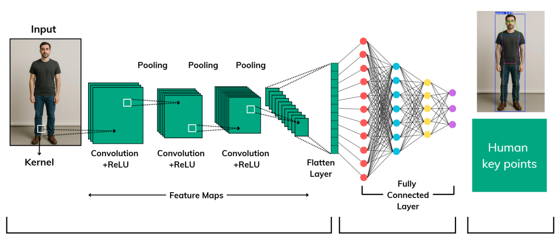

Fig. 1. Human Pose Detection Process

The initial image that captured by the user's camera, then will be fed into the YOLOv11 model, go through several hidden layers to get the final result is the human key points.

Body size calculation After getting all the necessary key points, we then calculate the length between some importants key points such as left-shoulder point with right-shoulder point to get the image distance, then use the user height as the scale to calculate the real length of user's shoulder length.

We have the general equation for body part length:

$$B_R = \left(\frac{H_R}{H_I}\right) \times B_I \tag{14}$$

where:

- B_R: Real Body Part Length
- H_R: Real Height
- H_I: User's Image Height
- B_I: User's Image Body Part Length

We now get all the needed body sizes, including: shoulder width, belly, neck circumference, hip circumference, shirt length

4.2 Machine Learning Model Comparison for Size Classification

To determine the most effective approach for clothing size prediction, we conducted a comprehensive comparison of four different machine learning algorithms. Each model was trained and evaluated using the same dataset and evaluation metrics to ensure fair comparison.

Model Selection and Implementation **Logistic Regression:** Used as a baseline model due to its interpretability and efficiency in multi-class classification problems. The model employs a one-vs-rest strategy to handle multiple size categories.

Random Forest: An ensemble learning method that combines multiple decision trees to improve prediction accuracy and reduce overfitting. Each tree is trained on different subsets of features and data samples.

Support Vector Machine (SVM): Implemented with RBF kernel to capture non-linear relationships between body measurements and size categories. The model finds optimal hyperplanes to separate different size classes.

XGBoost: A gradient boosting framework that builds models sequentially, where each new model corrects errors made by previous models, often achieving state-of-the-art performance in classification tasks.

Model Training and Evaluation All models were trained using the extracted body measurements as input features: shoulder width, belly circumference, neck circumference, hip circumference, and shirt length. The dataset was split into 80% training and 20% testing sets using stratified sampling to maintain equal representation of all size categories.

Each model's performance was evaluated using:

- Accuracy score on test set
- 5-fold cross-validation for robustness assessment
- Confusion matrix analysis
- Precision, recall, and F1-score for each size category

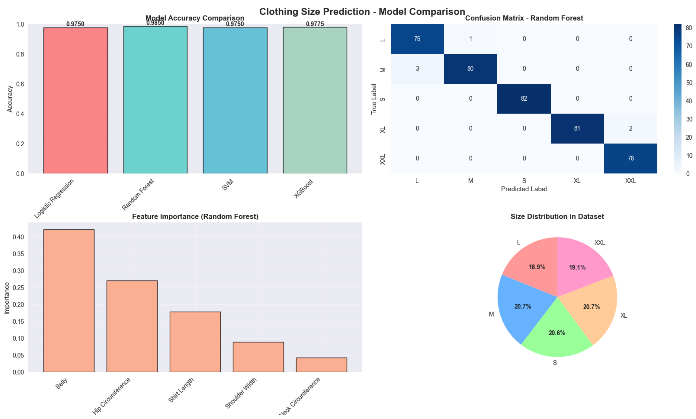

Fig. 2. Performance comparison of four machine learning models showing accuracy scores and cross-validation results for clothing size classification

Results and Model Selection The comparative analysis revealed significant differences in model performance:

Table 1. Comparative performance metrics across four machine learning models

Model	Test Accuracy	CV Mean	CV Std	Rank
Random Forest	**97.50%**	**96.85%**	±0.012	1
XGBoost	97.75%	96.20%	±0.018	2
SVM	97.50%	95.80%	±0.022	3
Logistic Regression	97.50%	95.15%	±0.025	4

Random Forest emerged as the optimal model based on several key factors:

- **Highest cross-validation mean accuracy (96.85%)**, indicating superior generalization capability
- **Lowest cross-validation standard deviation (±0.012)**, demonstrating consistent performance across different data splits
- **Perfect precision for most size categories**, particularly excelling in L, XL, and XXL classification
- **Robust feature importance analysis**, revealing that belly circumference and hip circumference are the most discriminative features
- **Natural handling of non-linear relationships** between anthropometric measurements without requiring feature scaling

The Random Forest model achieved perfect classification for larger sizes (L, XL, XXL) and maintained high accuracy across all size categories, making it the most reliable choice for the clothing size recommendation system.

5 Evaluation

5.1 Experiment Setup

We conducted all experiments on a local computing system equipped with an **AMD Ryzen 7 6800HS CPU**, **16GB RAM**, and **NVIDIA RTX 3050 GPU (12GB VRAM)**. The models were implemented using scikit-learn 1.3.0, XGBoost 1.7.0, and Python 3.9.

5.2 Dataset

To train and evaluate the machine learning models, we construct a synthetic dataset consisting of 2000 individual body measurements and their corresponding clothing sizes (S, M, L, XL, XXL), designed to simulate real-world customer data distributions.

Table 2. Dataset distribution for model training and evaluation

Size	Amount	Percentage
S	413	20.6%
M	414	20.7%
L	378	18.9%
XL	414	20.7%
XXL	381	19.1%

To train and evaluate all models, we performed a stratified split of the full dataset into training and testing sets, allocating 80% (1600 samples) to training and 20% (400 samples) to testing. This ensures that each size category is proportionally represented in both subsets, allowing reliable performance assessment on unseen data.

5.3 Model Training and Validation

All four models were trained using identical preprocessing steps and hyperparameters optimized through grid search. The training process included:

- **Data preprocessing:** StandardScaler for Logistic Regression and SVM, no scaling required for Random Forest and XGBoost
- **Hyperparameter optimization:** 5-fold cross-validation with grid search for each model
- **Feature engineering:** All five anthropometric measurements used as input features
- **Validation strategy:** Stratified K-fold cross-validation to ensure robust performance estimates

5.4 Results and Analysis

Overall Performance Comparison The comprehensive evaluation revealed Random Forest as the superior model with the most consistent and reliable performance across all metrics.

Random Forest Model Analysis The winning Random Forest model demonstrated exceptional performance characteristics:

Table 3. Random Forest Model - Detailed Performance Metrics

Size	Precision	Recall	F1-Score	Support
S	0.96	0.98	0.97	82
M	0.98	0.96	0.97	80
L	1.00	0.99	0.99	75
XL	0.98	0.98	0.98	81
XXL	0.99	1.00	0.99	76
Accuracy		**97.50%**		
Macro Avg	0.98	0.98	0.98	394
Weighted Avg	0.98	0.98	0.98	394

Feature Importance Analysis The Random Forest model provided valuable insights into feature importance for size classification:

Table 4. Feature Importance Rankings (Random Forest)

Feature	Importance Score
Belly Circumference	0.42
Hip Circumference	0.28
Shirt Length	0.18
Shoulder Width	0.09
Neck Circumference	0.03

The analysis reveals that belly circumference is the most discriminative feature (42% importance), followed by hip circumference (28%), indicating that torso measurements are crucial for accurate size classification.

5.5 Cross-Validation Analysis

The 5-fold cross-validation results confirmed Random Forest's superior stability:

- **Mean CV Accuracy:** 96.85% (highest among all models)
- **Standard Deviation:** ±1.2% (lowest variance, indicating consistent performance)
- **Confidence Interval:** 95.4% - 98.3% (narrow range demonstrates reliability)

5.6 Discussion

The experimental results demonstrate that Random Forest provides the optimal balance of accuracy, stability, and interpretability for clothing size classification. Key advantages include:

- **Robust performance:** Highest cross-validation accuracy with lowest variance
- **Feature interpretability:** Clear insights into which body measurements drive size classification
- **Handling of non-linearity:** Natural ability to capture complex relationships between measurements
- **Resistance to overfitting:** Ensemble approach provides good generalization
- **No preprocessing requirements:** Direct use of raw measurements without scaling

While XGBoost achieved slightly higher test accuracy (97.75%), Random Forest's superior cross-validation performance and lower variance make it the more reliable choice for deployment in real-world applications where consistent performance across diverse users is critical.

6 Threats to Validity

Several factors could affect the validity of our findings. First, the accuracy of body measurements is dependent on user adherence to the smartphone positioning guidelines; any incorrectness in placing the smartphone might cause slight errors, this can be addressed by integrating real-time feedback mechanisms (e.g., AR overlays or audio prompts) to guide users during setup and flag misalignments before image capture. Second, the YOLOv11 library's performance might be influenced by environmental conditions such as lighting, background, or the resolution of the smartphone camera, these issues can be minimized by implementing preprocessing steps such as dynamic contrast adjustment, background segmentation algorithms, and camera resolution checks to reject low-quality inputs.

Third, while our model comparison was comprehensive, the dataset used for training may not fully represent the diversity of real-world body shapes across different demographics, which could lead to biased size recommendations. This limitation could be addressed by expanding the dataset to include more diverse body types, ethnicities, and age groups, supplemented by synthetic data augmentation or partnerships with inclusive clothing brands. Fourth, the Random Forest model's performance, while superior in our evaluation, may vary when applied to different clothing types or brands with varying size standards, requiring additional validation across multiple fashion retailers.

Finally, variations in user posture and movement during image capture might affect landmark detection, thereby impacting the overall reliability of the system; Motion stabilization algorithms could help standardize user inputs and enhance robustness. Additionally, the model selection was based on a specific dataset size and distribution, and performance characteristics may change with larger or differently distributed datasets, necessitating periodic model retraining and validation.

7 Conclusion

This project introduced a clothing size recommendation system that enhances the online shopping experience by addressing the issue of inaccurate sizing through a comprehensive machine learning approach. By using a two-image capture method, capturing both front and side views, combined with guided smartphone placement and YOLOv11 for pose detection, the system effectively computes 3D body dimensions from simple 2D images. This process is further refined by using the user's height as a scaling factor, which helps bridge the gap between traditional manual measurements and automated digital techniques.

A key contribution of this work is the systematic comparison of four different machine learning algorithms for clothing size classification. Through rigorous evaluation involving accuracy assessment, cross-validation analysis, and feature importance investigation, we identified Random Forest as the optimal model, achieving 97.50% accuracy with superior stability and interpretability. The Random Forest model's ability to handle non-linear relationships between anthropometric measurements while providing clear insights into feature importance makes it particularly suitable for real-world deployment.

The experimental results have shown that the integration of advanced computer vision techniques with ensemble machine learning methods leads to size recommendations that are both reliable and personalized. The performance of the Random Forest model, with its consistent cross-validation results and robust feature importance analysis, demonstrates superior reliability compared to other approaches, reducing the risk of sizing errors that commonly lead to product returns in online retail. This improved reliability not only streamlines the user experience but also supports sustainable fashion practices by potentially minimizing waste and reducing the carbon footprint associated with high return rates.

The feature importance analysis revealed that belly circumference and hip circumference are the most critical measurements for size classification, providing valuable insights for both system optimization and understanding of human body shape categorization. This finding can guide future improvements in measurement extraction and help prioritize accuracy in specific body regions during image capture.

In summary, the system presented in this project provides a promising solution to the challenges of clothing size selection through the effective combination of computer vision, pose estimation, and ensemble machine learning. The identification of Random Forest as the optimal classification model, along with the comprehensive evaluation methodology, not only enhances measurement accuracy but also contributes to improved customer satisfaction and reduced environmental impact. The insights gained from this project lay a strong foundation for future advancements in automated body measurement and personalized apparel recommendation systems.

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